

Power signal

1. Energy

- Total normalized energy is finite

Normalized average power is finite & non zero

- Energy is obtained by,

$$E = \int_{-\infty}^{\infty} x^2(t) dt$$

(Avg) Power is obtained by

$$P_{av} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t) dt$$

- Non periodic signals are generally periodic signals

Periodic signals are periodic signals

- These signals are time limited

signals can exist over time

- Power of energy signal is zero

Energy of power signal is finite

- eg: $0.5e^{-2} u(n)$

eg: $\sin(0.2\pi n)$

$$x[n] = e^{j\left(\frac{\pi n}{2} + \frac{4\pi}{7}\right)}$$

Here: complex signal.

so,

$$E = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2$$

$$= \lim_{N \rightarrow \infty} \sum_{n=-N}^N |e^{j\left(\frac{\pi n}{2} + \frac{4\pi}{7}\right)}|^2$$

$$= \lim_{N \rightarrow \infty} \sum_{n=-N}^N 1 = \lim_{N \rightarrow \infty} 2N+1$$

$$= \infty$$

and average power,

$$P_{av} = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \times 2N+1$$

$$= \lim_{N \rightarrow \infty} 1$$

$$= 1$$

Since $E = \infty$ (finite infinite)

$P = 1$ (finite non zero)

so power signal.

8.2 Ba

1. $y[n] = x[n] + x[-n]$

(i) causal:

at $n=0$

$$y[0] = x[0] + x[0]$$

$$y[1] = x[1] + x[-1]$$

$$y[2] = x[2] + x[-2]$$

⋮

since the output depends on $k=n$ and $k \leq n$, the system is causal.

(ii) Time invariant

$$y[n] = x[n] + x[-n]$$

adding a delay in y : $y[n-n_0] = x[n-n_0] + x[-(n-n_0)]$
and, $= x[n-n_0] + x[-n+n_0]$

for delay in x : $x_1 = x[n-n_0]$

$$y_1 = x_1[n] + x_1[-n]$$

$$= x[n-n_0] + x[-n-n_0], y, \neq y_1 \text{ so invariant}$$

1) 8L Bh

$$1. x[n] = \cos\left(\frac{2\pi n}{5}\right) + \sin\left(\frac{\pi n}{3}\right)$$

To be periodic: LCM of 2 ^{Time periods} frequencies should exist, of summed sinusoids.

$$\text{so, } f_1 = \frac{2\pi}{5}, \text{ so } T = 5$$

$$f_2 = \frac{\pi}{3}, \text{ so } T = 3 \times 2 = 6$$

LCM is $T = 30$, so, it exists. and thus, $x[n]$ is periodic.
and fundamental period is 30.

8L Ba

$$1. y[n] = x[n] + e^a y[n-1]$$

Taking Z-transform,

$$Y(z) = X(z) + e^a z^{-1} Y(z)$$

$$\frac{Y(z)}{X(z)} = \frac{1}{1 - e^a z^{-1}}$$

$$\text{or, } H(z) = \frac{1}{1 - e^a z^{-1}}$$

$$\therefore h[n] = (e^a)^n u[n]$$

BIBO stable:

// by Paley - Wiener criteria:
system is stable if

necessary & sufficient condition:

$|h[n]|$ sum of over time is finite.

$$\text{i.e. } \lim_{N \rightarrow \infty} \sum_{n=-N}^{n=N} |h[n]|$$

$$N \rightarrow \infty \quad n = -N$$

$$= \lim_{N \rightarrow \infty} \sum_{n=-N}^{n=N} |(e^a)^n u[n]|$$

$$= \lim_{N \rightarrow \infty} \sum_{n=0}^N |(e^a)^n|$$

This is absolutely summable if e^a is decreasing over time.
ie. a must be less than 0.

$\therefore$ system is stable if $a < 0$, or else not stable for $a \geq 0$.

80 Bh

$$1. y[n] = \sum_{k=0}^n x[k]$$

memory:

$$\begin{aligned} y[n] &= \sum_{k=0}^n x[k] \\ &= x[0] + x[1] + \dots + x[n] \end{aligned}$$

ie. $y[n]$ depends on values other than $x[n]$, so it is ^{not} memoryless.

time invariant:

if delay in output:

$$y[n] = \sum_{k=0}^n x[k]$$

$$\text{ie. } y[n-n_0] = \sum_{k=0}^{n-n_0} x[k]$$

if delay in input: $x_1[n] = x[n-n_0]$:

$$y_1[n] = \sum_{k=-\infty}^n x_1[k]$$

$$= \sum_{k=-\infty}^n x[k-n_0]$$

Take $k - n_0 = k_1$, or, $k = k_1 + n_0$

at $k = -\infty$, $k_1 = -\infty$

$k = \infty$, $k_1 = \infty - n_0$

so,

$$y_1[n] = \sum_{k_1=-\infty}^{n-n_0} x[k_1]$$

where $y[n] = y_1[n]$, is the invariant.

stable:

$$y[n] = \sum_{k=0}^{\infty} x[k]$$

for a bounded input $x[k]$, say, $x[k]$,

we have: $x[n] = 1$ for $n \geq 0$ i.e. bounded.

so,

$$y[n] = \sum_{k=0}^n 1$$

$$= n + 1$$

as $n \rightarrow \infty$, $y[n] \rightarrow \infty$

so $y[n]$ is not bounded.

80 Ba

L a) $x[n] = \sin(n\pi) + \cos(n\pi)$

$$T_1 = 2 \times 1 = 2$$

$$T_2 = 2 \times 1 = 2$$

LCM of T_1 & $T_2 = 2$, so periodicity = 2 as LCM exists.

$$b) x[n] = \sin\left[\frac{3n\pi}{5}\right] + \cos\left[\frac{4n\pi}{7}\right]$$

$$T_1 = 2 \times \frac{5}{3} = \frac{10}{3}$$

$$T_2 = 2 \times \frac{7}{4} = \frac{7}{2}$$

LCM valⁿ T_1 & T_2 : $\frac{21}{2}$ 70 $\left(\frac{10}{3} \times 3 \times 7, \frac{7}{2} \times 2 \times 5\right)$ as integer

since LCM exists, periodic signal
so period = 70

79 Bn

Energy signals are those signals that have finite energy & zero power.

Power signal are those signals that have infinite energy & finite power.

$$x[n] = \cos\left[\frac{2\pi n}{5}\right] + \sin\left[\frac{\pi n}{3}\right]$$

$$T_1 = 5$$

$$T_2 = 3 \times 2 = 6$$

since LCM exists, = 30

$\therefore$ periodic, period = 30

78 Bh

$$x[n] = \cos\left[\frac{\pi n}{2}\right] \cos\left[\frac{\pi n}{4}\right]$$

$$= \frac{1}{2} \left[\cos\left[\frac{\pi n}{2} + \frac{\pi n}{4}\right] + \cos\left[\frac{\pi n}{2} - \frac{\pi n}{4}\right] \right]$$

$$= \frac{1}{2} \left[\cos\left[\frac{3\pi n}{4}\right] + \cos\left[\frac{\pi n}{4}\right] \right]$$

$$T_1 = 2 \times \frac{4}{3} = \frac{8}{3}$$

$$T_2 = 2 \times 4 = 8$$

LCM exists,

periodic, period = LCM = 8

79 Ba

→ 82 Bh, pg 1

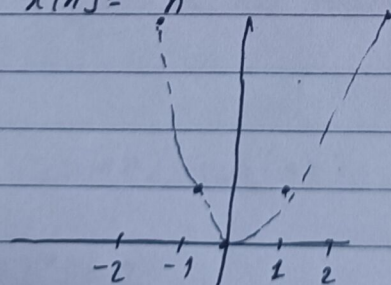
78 Bh

→ 78 B

76 Ch

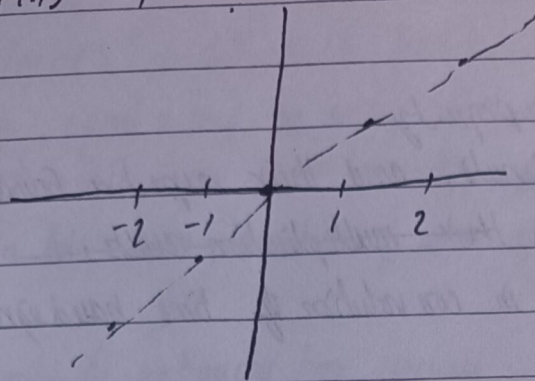
d.T.

even type, signals are those signals that are symmetric about origin.

e.g. $x[n] = n^2$ 

odd type signals are those signals that are anti-symmetric about origin.

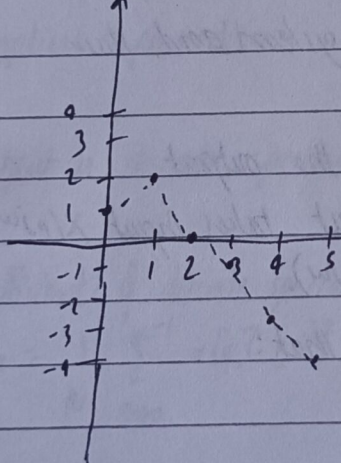
$$x[n] = n$$



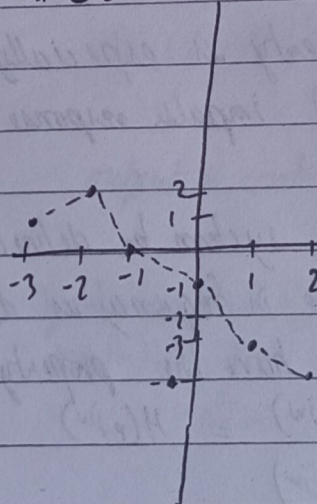
Given:

$$x[n] = \{2, 2, 0, -1, -3, -4\}$$

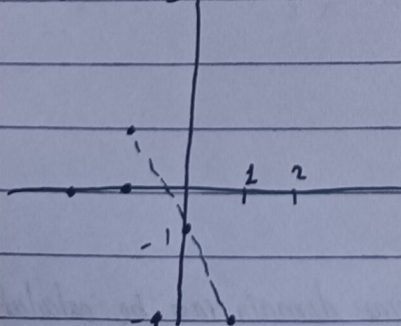
so, $x[n]$:



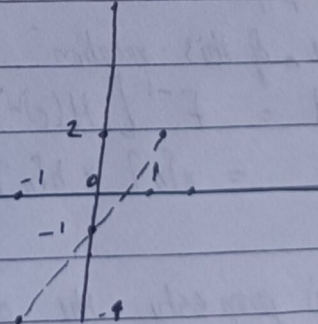
$x[2n]$ $x[n+3]$



$x[2n+3]$



$x[-2n+3]$



H 76 Ash

Fourier Transform multiplication property:

If x and y are two time signals, and their respective Fourier transforms are $X(e^{j\omega})$ and $Y(e^{j\omega})$. Then their multiplication results in their FT's multiplication results in convolution of time-based signals. i.e.:

$$\text{if } x[n] \xrightarrow{FT} X(e^{j\omega})$$

$$\text{and } y[n] \xrightarrow{FT} Y(e^{j\omega})$$

then,

$$X(e^{j\omega}) \cdot Y(e^{j\omega}) \xrightarrow{FT} x[n] * y[n]$$

This property is especially useful for LTI system and their analysis of impulse response.

as the output
i.e. let a system be defined as $Y(e^{j\omega})$ that takes input $X(e^{j\omega})$. The response in frequency is defined by $H(e^{j\omega})$.

Then, we have the property of LTI system that:

$$Y(e^{j\omega}) = H(e^{j\omega}) X(e^{j\omega})$$

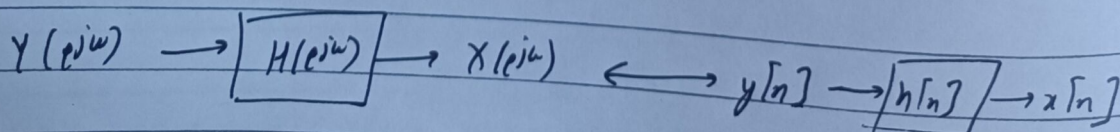
$$X(e^{j\omega})$$

$$\text{or, } Y(e^{j\omega}) = H(e^{j\omega}) \cdot X(e^{j\omega})$$

Taking inverse of this relation:

$$y[n] = F^{-1} [H(e^{j\omega}) \cdot X(e^{j\omega})]$$
$$= x[n] * h[n]$$

Thus, by this property, the output in real-time domain can be calculated based on the convolution of x and h :



~~Parasat~~

Dirichlet's conditions for Fourier Series:

- i. any signal $x[n]$ has to be periodic, finite and single-valued
 - ii. ~~that~~ $x[n]$ has finite number of discontinuities in any one period.
 - iii. $x[n]$ has finite number of maxima and minima.
- only for continuous time signals.

75 Ch

Fourier Series

- Applies to periodic
- output is discrete spectrum
- coefficients of discrete spectrum given by:

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j^k \left(\frac{2\pi}{N}\right) n}$$

- special case for Fourier Transform, as samples for FT, of periodic signal

Fourier Transform

Applies to non-periodic signals.

output is continuous spectrum.

Spectrum given by

$$X(j\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

general case for signals.

7583

$$y[n] = x^2[n]$$

$$y[n] = \cos\left[\frac{5\pi n}{8} + \frac{\pi}{9}\right]$$

a) let:

$$y_1[n] = x_1^2[n]$$

$$y_2[n] = x_2^2[n]$$

and,

$$ay_1[n] + by_2[n] = ax_1^2[n] + bx_2^2[n]$$

$$\text{let } x_3[n] = ax_1^2[n] + bx_2^2[n]$$

$$y_3[n] = x_3^2[n]$$

$$= a^2 x_1^2[n] + b^2 x_2^2[n] + 2ab x_1 x_2$$

$$+ 2ab x_1[n] x_2[n]$$

$$\neq ay_1[n] + by_2[n]$$

so not linear system

b) let

$$y_1[n] = \cos$$

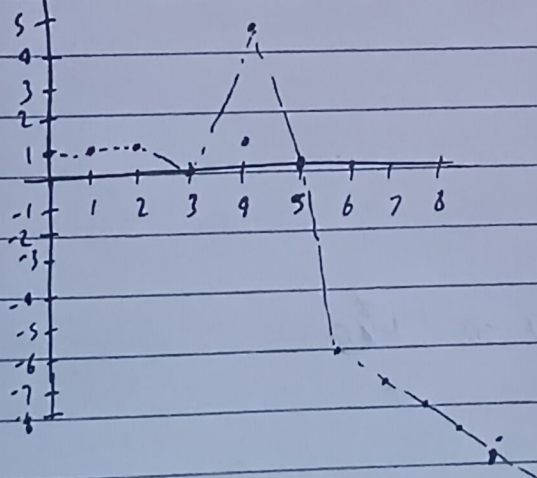
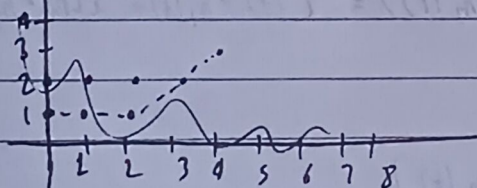
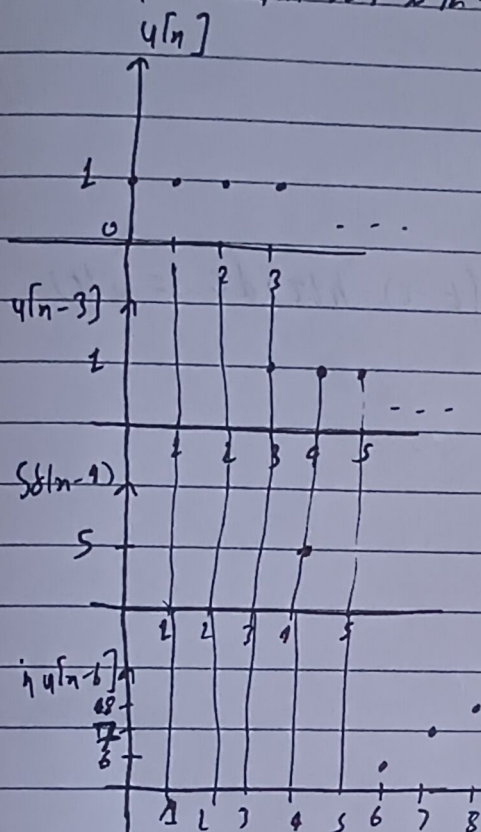
Here, $y[n] = \cos\left[\frac{5\pi n}{2} + \frac{\pi}{9}\right]$ is a function on n -domain and

does not act on any sequence i.e. $y[n]$ is a sequence itself, thus it cannot be classified as a linear system.

But, if classified as a sequence, it is not linear sequence, rather periodic sequence.

746h

$$x[n] = u[n] - u[n-1] + 5\delta[n-4] - 7u[n-6]$$



Properties of LTI system

1. Commutative property,

$$\text{if } x(t) * h(t) = y(t)$$

$$\text{then } h(t) * x(t) = y(t)$$

$$\text{as in: } \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau = y(t)$$

2. Distributive property:

$$\text{if } y_1 = x(t) * h_1(t)$$

$$\text{and } y_2 = x(t) * h_2(t)$$

then,

$$y = x(t) * h(t)$$

$$\text{where, } y = y_1 + y_2$$

$$h(t) = h_1(t) + h_2(t)$$

$$\text{i.e. } y(t) = x(t) * (h_1(t) + h_2(t)) = x(t) * h_1(t) + x(t) * h_2(t)$$

3. Associative property:

$$\text{if } y(t) = x(t) * [h_1(t) * h_2(t)]$$

$$\text{then } y(t) \text{ is also } = [x(t) * h_1(t)] * h_2(t)$$

4. Static / dynamic LTI system

↓

memoryless

system is memoryless if $h(t) = 0$; $t \neq 0$

$$\text{i.e. } y = kx(t)$$

$$\text{and } h(t) = k\delta(t)$$

5. Invertibility

LTI system is invertible if it has LTI inverse $h_1[n]$ such that

$$\text{for } y[n] = h[n] * x[n]$$

$$h[n] * h_1[n] = \delta[n]$$

If invertible, system's output = input:

$$x[n] \rightarrow |h[n]| \rightarrow |h_1[n]| \rightarrow x[n] = y[n]$$

6. Stability

if $\sum_{n=-\infty}^{\infty} h[n]$ is bounded, then the output of system is bounded
(i.e. $< \infty$) for bounded input.

7. Causality

if $h[n] = 0$; $n < 0$

then system is causal.

2. Ex:

$$a) y[n] = y[n-4] + x[n-4]$$

$$\text{or, } Y(z) = z^{-4} Y(z) + z^{-4} X(z)$$

$$\text{or, } \frac{Y(z)}{X(z)} = H(z) = \frac{z^{-4}}{1 - z^{-4}}$$

$$\text{let } x_1 = x[n-n_0]$$

$$\text{so, } y_1[n] = y_1[n-4] + x_1[n-4]$$

$$= y_1[n-4] + x[n-n_0-4]$$

now, adding delay in output,

$$y[n-n_0] = y[n-n_0-4] + x[n-n_0-4]$$

$$= y_1[n-n_0-4] + x[n-n_0-4]$$

$$= y_1[n-4] + x[n-n_0-4]$$

$\therefore y_1[n] = y[n-n_0]$, so time invariant.

b) $y[n] = x^2[n]$

if b/w

let $y_1[n] = ax_1^2[n]$

$y_2[n] = bx_2^2[n]$

then, let

$y_3[n] = ax_1^2[n] + bx_2^2[n]$

and,

$x_3[n] = ax_1[n] + bx_2[n]$

then, we have,

$y_3[n] = (x_3[n])^2$

$= a^2x_1^2[n] + b^2x_2^2[n] + 2abx_1[n]x_2[n]$

$\neq ax_1^2[n] + bx_2^2[n]$

Thus, not linear.

79 Ach

two signals

73 shr, 72ch

fourier series coefficients,

let $x[n]$ be signal, then,

fourier series coefficients a_k :

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j k \frac{2\pi}{N} n}$$

depending upon the type of signal $x[n]$, we may calculate these separately.



if $x[n]$ is a functional / descriptive / analytical series:

e.g.: $\sin()$, power series, etc.

we do analytical approach by solving the sum

$$\sum x[n] e^{-jk \frac{2\pi}{N} n} \quad \sum a_k e^{-jk \left(\frac{2\pi}{N}\right) n}$$

into a new function

else, if $x[n]$ is descriptive,

e.g.: $\{1, 0, 1, 0, 0, 2\}$, $\sum x[n] e^{-jk \frac{2\pi}{N} n}$

then we may simply perform the sum manually to derive a function from the sum.

(3 marks, so not doing a long example)

(ideally what short example can be)

74 Ch + example 11.

for analytical function: $x = \sin(\omega_0 n)$

$$\text{so, } x[n] = e^{j\omega_0 n} - e^{-j\omega_0 n}$$

$$= \frac{1}{2j} e^{j \left(\frac{2\pi}{N}\right) n} - \frac{1}{2j} e^{-j \left(\frac{2\pi}{N}\right) n}$$

so, we already have the $\sin$ as a series of complex numbers, i.e.:

$$x[n] = \sum_{k=0}^{N-1} a_k e^{-jk \left(\frac{2\pi}{N}\right) n}$$

$$\text{Thus, at } k=1: a_k = \frac{-1}{2j}$$

$$\text{at } k=-1, a_k = \frac{1}{2j}$$

, thus, coefficients are found by analysis.

72 xa
 $x[n] = u[n]$

Here,

$$E = \sum_{n=0}^{\infty} x[n]$$

$$= \sum_{n=0}^{\infty} 1$$

$$= \infty$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^N x[n]$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^N 1$$

$$= \lim_{N \rightarrow \infty} \frac{N}{2N+1}$$

$$= \frac{1}{2}$$

no power type.

$$x[n] = \delta[n]$$

$$E = \lim_{N \rightarrow \infty} \sum_{n=-N}^N x[n]$$

$$= \lim_{N \rightarrow \infty} 1$$

$$= 1$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x[n]$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^0 \delta[n]$$

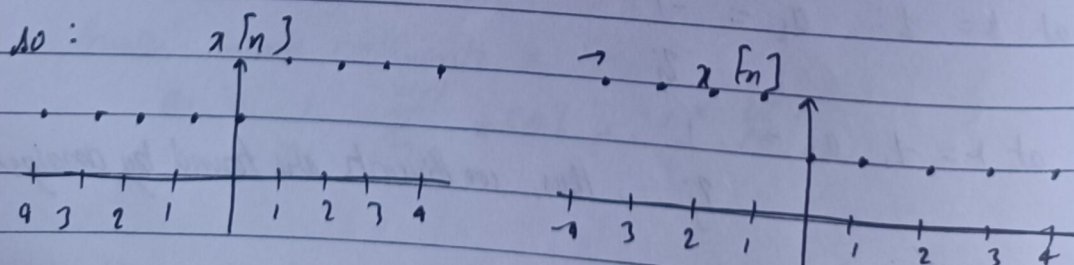
$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \times 1$$

$$= 0$$

$\therefore$ energy signal

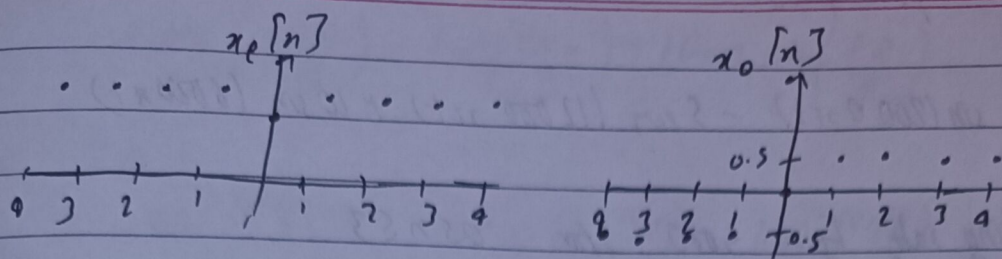
71 ch: / 71 Shv

$$x[n] = \begin{cases} 1 & \text{for } -4 \leq n \leq 0 \\ 2 & \text{for } 1 \leq n \leq 4 \end{cases}$$



so, even signal: $\frac{x[n] + x[-n]}{2} : x_e[n]$

odd signal: $\frac{x[n] - x[-n]}{2} : x_o[n]$



70 ch

$$\cos\left(\pi \frac{n^2}{8}\right)$$

function is not periodic as input of cosine is not linear (n^2)

$$\cos\left(\frac{n}{2}\right) \cos\left(\frac{\pi n}{4}\right)$$

$$= \frac{1}{2} \left[\cos\left(\frac{n}{2} + \frac{\pi n}{4}\right) + \cos\left(\frac{n}{2} - \frac{\pi n}{4}\right) \right]$$

$$= \frac{1}{2} \left[\cos\left(\left(\frac{1}{2} + \frac{\pi}{4}\right)n\right) + \cos\left(\left(\frac{1}{2} - \frac{\pi}{4}\right)n\right) \right]$$

$$= \frac{1}{2} \left[\cos\left(\frac{2+\pi}{4}n\right) + \cos\left(\frac{2-\pi}{4}n\right) \right]$$

since the angular frequency is not a ^{integer} multiple of π , either of the 2 sinusoids

both sinusoids have angular frequency that is not an ^{rational} integer multiple of π , so can't be expressed in terms of $\frac{M}{N}\pi$, so not periodic.

Sample type question.

8014

$$x(t) = \sin(2000\pi t) - 5\cos(12000\pi t) + 10\sin(6000\pi t)$$

sampling rate, $F_s = 5000$, for $0 \leq n \leq 3$

Here,

$$\Omega_1 = 2000\pi \rightarrow f_1 = 1000\text{ Hz}$$

$$\Omega_2 = 12000\pi \rightarrow f_2 = 6000\text{ Hz}$$

$$\Omega_3 = 6000\pi \rightarrow f_3 = 3000\text{ Hz}$$

largest frequency: 6000 Hz , so Nyquist rate = 12000 min^m .

since sampling at $\frac{n}{5000} = t$:

$$\omega_1 = \frac{2000\pi}{5000} = \frac{2\pi}{5}$$

$$\omega_2 = \frac{12000\pi}{5000} = \frac{12\pi}{5}$$

$$\omega_3 = \frac{6000\pi}{5000} = \frac{6\pi}{5}$$

for aliasing:

$$f_1 < \frac{F_s}{2} \text{ no aliasing}$$

$$f_2 > \frac{F_s}{2}, \text{ aliasing} \rightarrow \omega_2' = \omega_2 - 2\pi = \frac{2\pi}{5}$$

↙ 1 cycle of frequency reduced.

$$f_3 > \frac{F_s}{2}, \text{ aliasing} \rightarrow \omega_3' = \omega_3 - 2\pi = -\frac{4\pi}{5}$$

so, after sampling,

$$x[n] = \sin\left(\frac{2\pi n}{5}\right) - 5\cos\left(\frac{2\pi n}{5}\right) - 10\sin\left(-\frac{4\pi n}{5}\right)$$

$$= \sin\left(\frac{2\pi n}{5}\right) - 5\cos\left(\frac{2\pi n}{5}\right) + 10\sin\left(\frac{4\pi n}{5}\right)$$

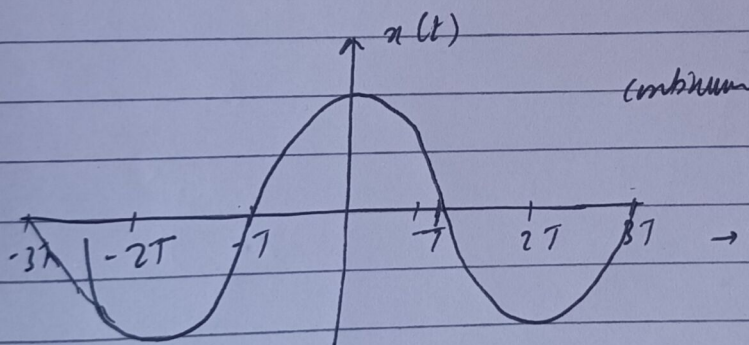
for 0 to 3:

n	$\sin(2\pi n/5)$	$-5\cos(2\pi n/5)$	$10\sin(4\pi n/5)$	x
0	0	-5	0	-5
1	0.9510	1.545	5.878	8.373
2	0.5878	4.045	-9.510	-4.872
3	-0.5878	4.045	9.510	12.967

Sampling is process of converting a continuous time signal $x(t)$ into a discrete-time signal $x[n]$ by taking its values only at discrete time instants, usually uniformly spaced,

$$x[n] = x[nT], \quad T = \frac{1}{f_s} \quad \text{sampling period.}$$

$f_s \leftarrow$ sampling frequency



$\rightarrow T$ is any arbitrary time period here

